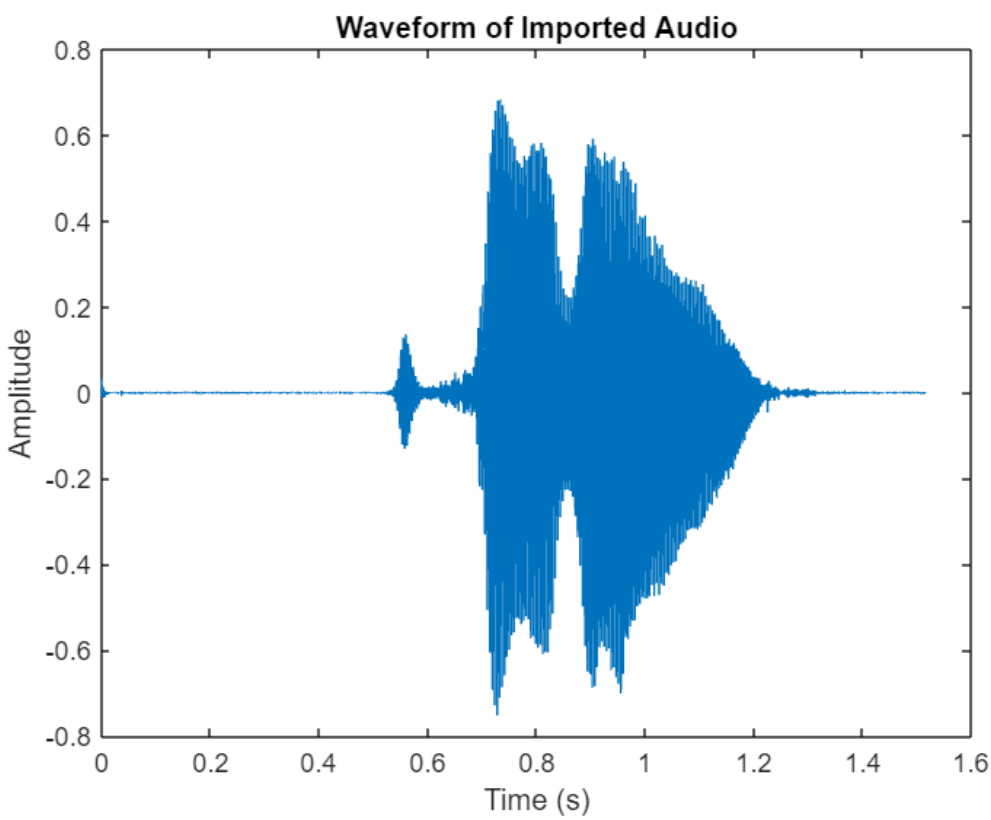
Load the Signal

```
[signal,fs]=audioread("s3.wav");
```

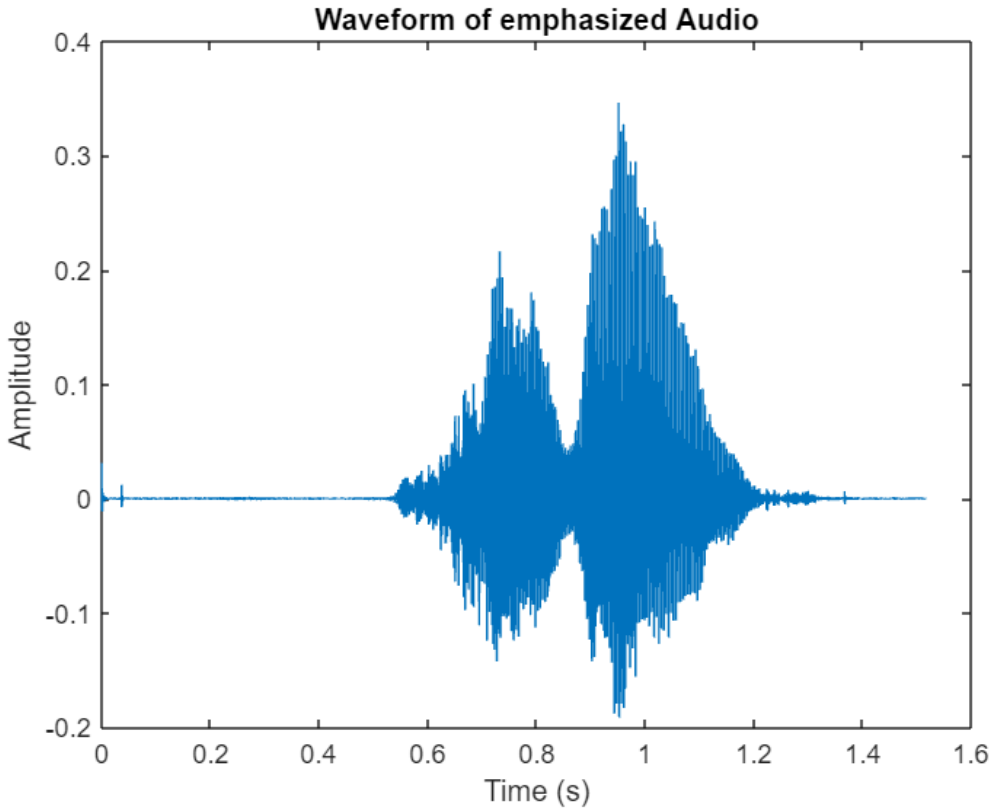
Visualizing the Signal

```
t = (0:length(signal)-1)/fs;   % Time axis
plot(t, signal);
xlabel('Time (s)');
ylabel('Amplitude');
title('Waveform of Imported Audio');
```



Preemphasising

```
alpha = 0.97;
emphasized = preemphasis(signal, alpha);
plot(t, emphasized);
xlabel('Time (s)');
ylabel('Amplitude');
title('Waveform of emphasized Audio');
```



Framing

```
frame_length = 256;
hop_size=100;
frames = frame_signal(emphasized, frame_length, hop_size);
frames
```

```
frames = 256×187
    0.0312    0.0013   -0.0005         0   -0.0005   -0.0005         0    0.0005 ···
   -0.0093    0.0004    0.0005         0    0.0005   -0.0010   -0.0005   -0.0005
   -0.0086    0.0004   -0.0005    0.0005         0    0.0005    0.0005    0.0005
    0.0062    0.0009   -0.0000   -0.0005   -0.0005         0         0         0
   -0.0048    0.0004   -0.0005    0.0010    0.0005         0         0   -0.0005
   -0.0060    0.0010   -0.0000   -0.0005         0   -0.0005    0.0005    0.0005
    0.0031    0.0005   -0.0000   -0.0005         0   -0.0000   -0.0005    0.0005
    0.0027    0.0010   -0.0000    0.0005         0   -0.0000         0   -0.0005
   -0.0060    0.0005   -0.0005   -0.0005   -0.0005   -0.0000         0         0
    0.0006    0.0001   -0.0000         0    0.0005    0.0005         0   -0.0005
      ⋮
```

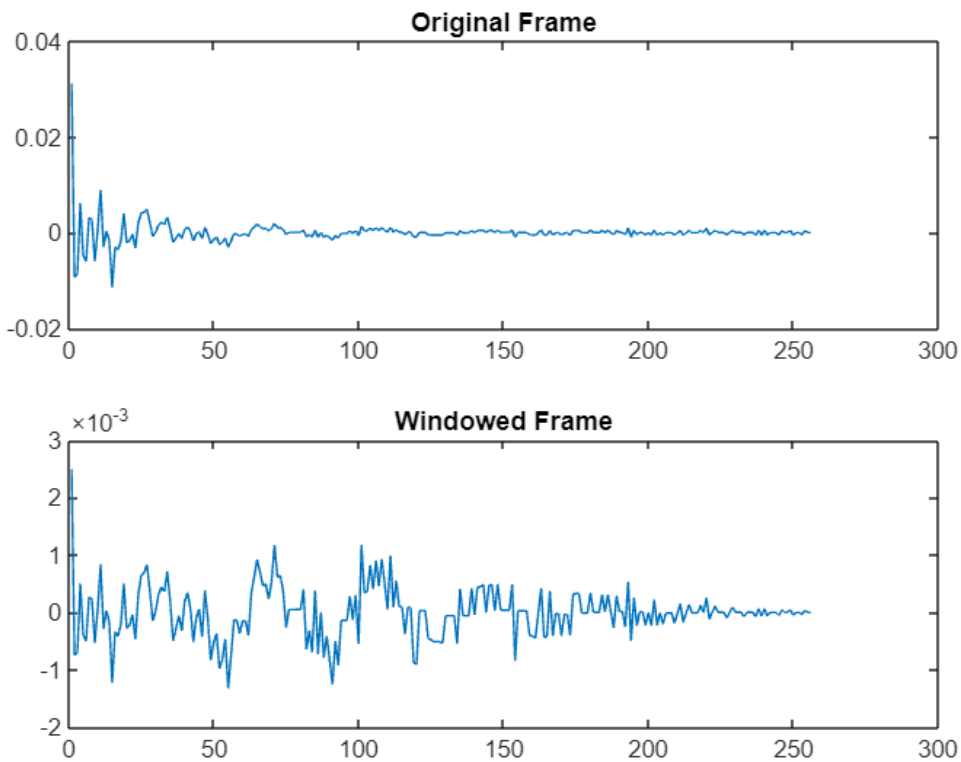
Applying Hamming Window

```
windowed_frames=apply_window(frames);
```

```
frameIdx = 1;
subplot(2,1,1);
```

```
plot(frames(:, frameIdx));
title('Original Frame');

subplot(2,1,2);
plot(windowed_frames(:, frameIdx));
title('Windowed Frame');
```




```
[frame_length_windowed,number_of_frames]=size(windowed_frames);
frame_length_windowed
```

```
frame_length_windowed = 256
```

```
number_of_frames
```

```
number_of_frames = 187
```

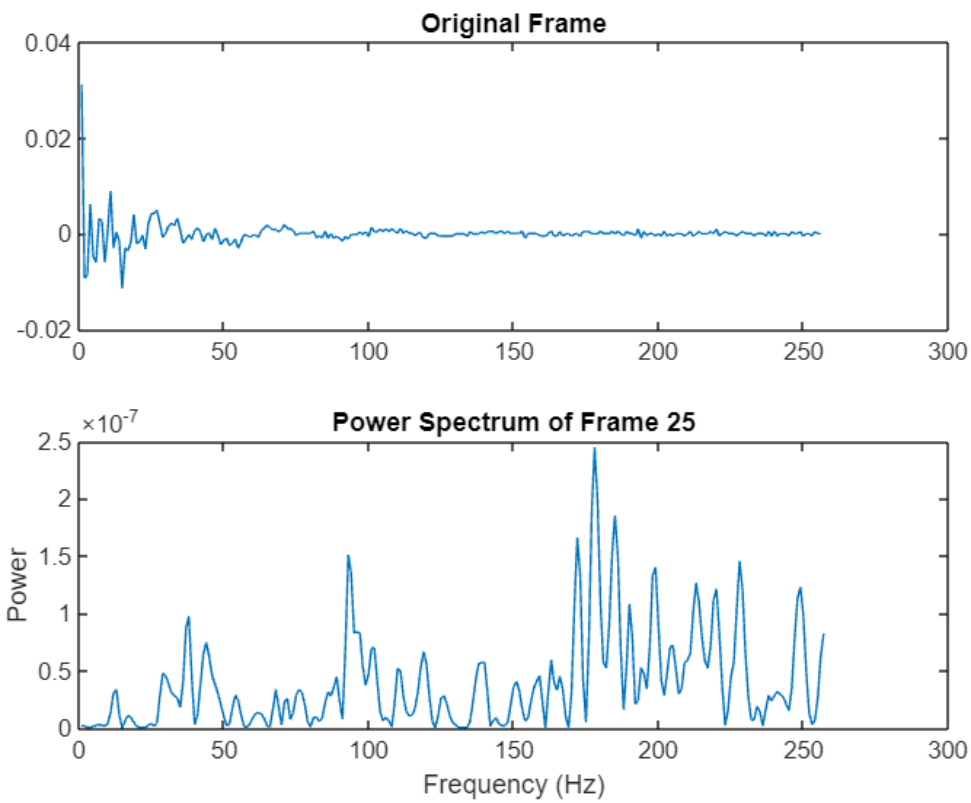
Compute the Power Spectrum

```
nfft = 512;
power_spectrum = compute_power_spectrum(windowed_frames, nfft)
```

```
power_spectrum = 257×187
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000 ···
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
```

```
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
      ⋮
```

```
frameIdx = 25;   % pick a frame
plot(power_spectrum(:, frameIdx));
xlabel('Frequency (Hz)');
ylabel('Power');
title(sprintf('Power Spectrum of Frame %d', frameIdx));
```



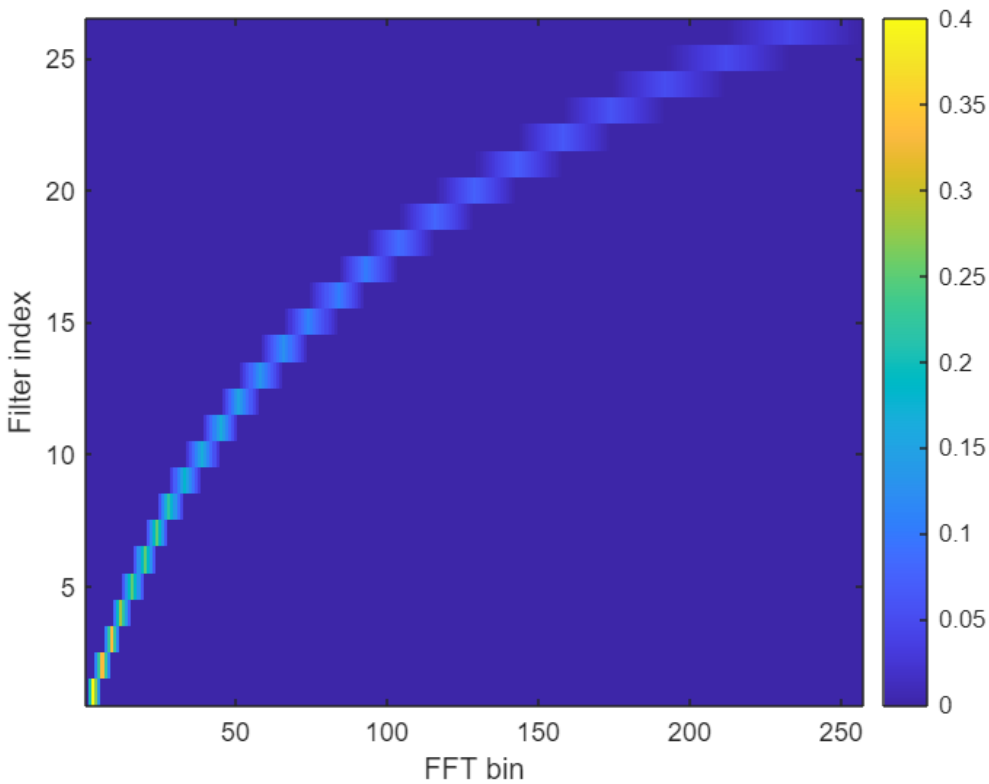
Creating Mel Filter Bank and Applying

```
Num_of_Mel_filters=26;
Melbank_filter=melfb(Num_of_Mel_filters,nfft,fs)
```

```
Melbank_filter = 26×257
         0    0.2000    0.4000    0.2667    0.1333         0         0         0 ···
         0         0         0    0.1111    0.2222    0.3333    0.2222    0.1111
         0         0         0         0         0         0    0.1111    0.2222
         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0
```

```
         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0
    ⋮
```

```
figure;
imagesc(Melbank_filter);
axis xy;
xlabel("FFT bin")
ylabel("Filter index")
colorbar;
```

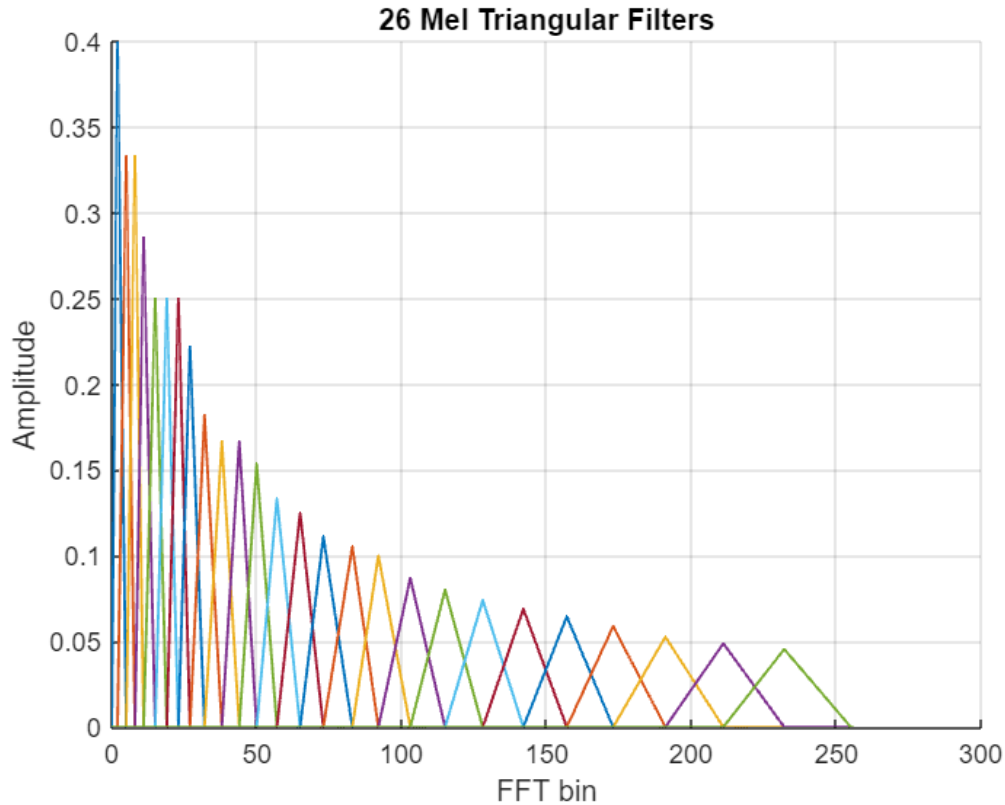



```
figure;
hold on;

[num_filters, num_bins] = size(Melbank_filter);

for m = 1:num_filters
    plot(0:num_bins-1, Melbank_filter(m,:), 'LineWidth', 1);
end

xlabel('FFT bin');
ylabel('Amplitude');
title('26 Mel Triangular Filters');
grid on;
hold off;
```



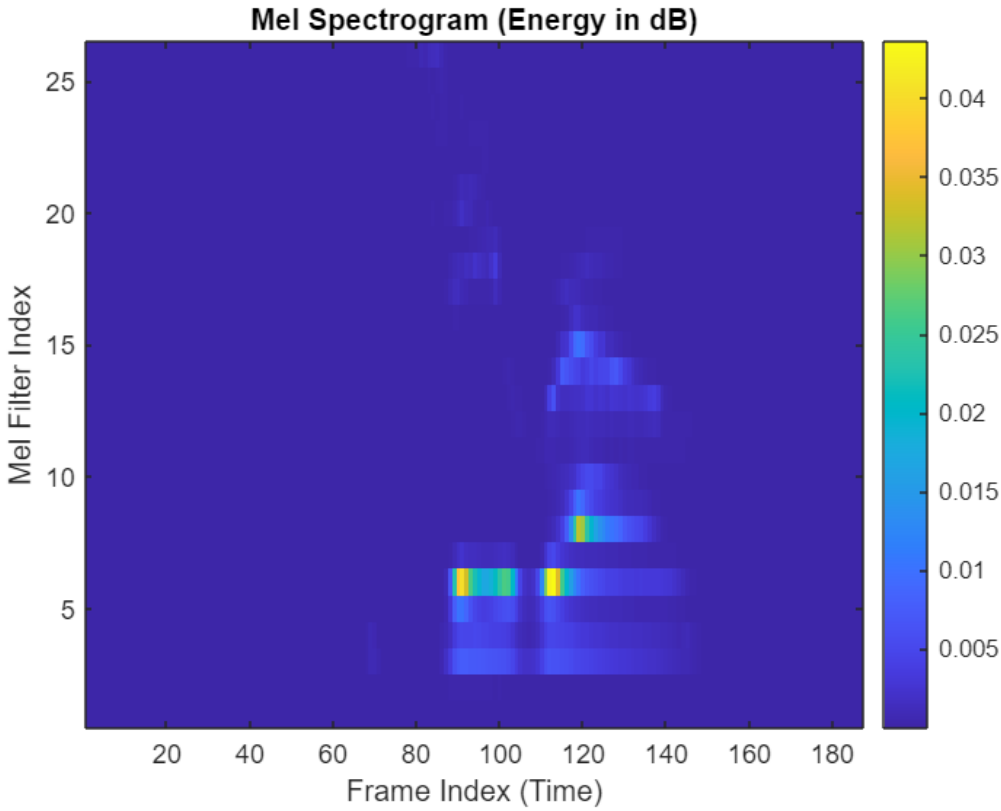

```
mel_energies = apply_mel_filterbank(power_spectrum, fs, nfft, num_filters)
```

```
mel_energies = 26×187
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000 ···
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
      ⋮
```

Mel Energy

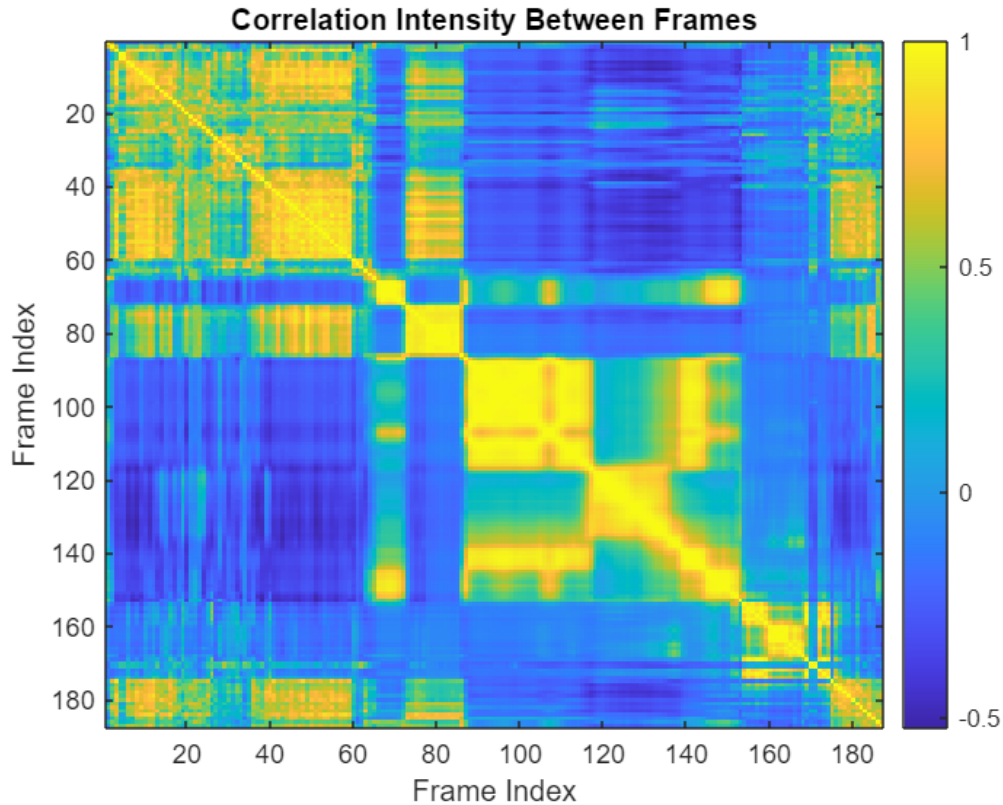
```
% Plot as an image
imagesc(mel_energies);

% Adjust axes
axis xy;   % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('Mel Filter Index');
title('Mel Spectrogram (Energy in dB)');
colorbar;
```



Correlation across Frames

```
R = corrcoef(mel_energies);   % correlation across frames
imagesc(R);
colorbar;
xlabel('Frame Index');
ylabel('Frame Index');
title('Correlation Intensity Between Frames');
```



Apply Log

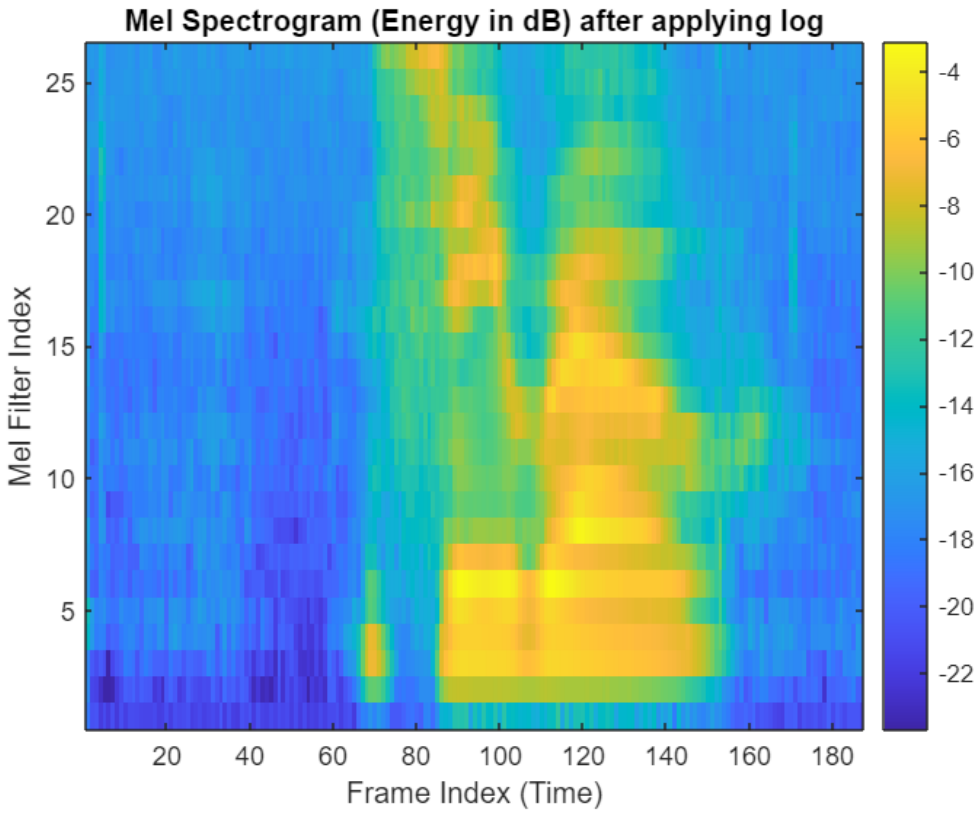
```
log_mel = apply_log(mel_energies)
```

```
log_mel = 26×187
  -18.5033  -20.7790  -20.5791  -21.9428  -21.1276  -22.0190  -20.8916  -22.1367 ···
  -17.4673  -21.3604  -21.6382  -22.7959  -23.6812  -23.6670  -23.4893  -21.9964
  -17.7948  -20.2586  -20.9840  -20.1582  -20.8246  -23.6149  -21.3135  -20.4045
  -13.9581  -16.5849  -18.3196  -18.0312  -18.0820  -20.1150  -19.4175  -19.6294
  -13.4247  -16.3370  -18.0255  -17.6427  -17.1409  -18.5406  -18.8897  -18.8388
  -16.6198  -19.7256  -19.9608  -17.8100  -18.1578  -19.0504  -19.9844  -19.2378
  -16.3879  -18.4325  -19.3966  -17.9174  -18.5720  -18.3069  -19.7267  -19.4056
  -15.8839  -19.1571  -19.1786  -16.6927  -16.9527  -19.8188  -19.2811  -19.9371
  -17.4075  -19.8105  -19.0898  -16.7304  -17.4547  -21.1375  -20.1152  -20.2067
  -17.5053  -19.0326  -18.8475  -16.9758  -18.7498  -18.2646  -18.1552  -18.4922
      ⋮
```

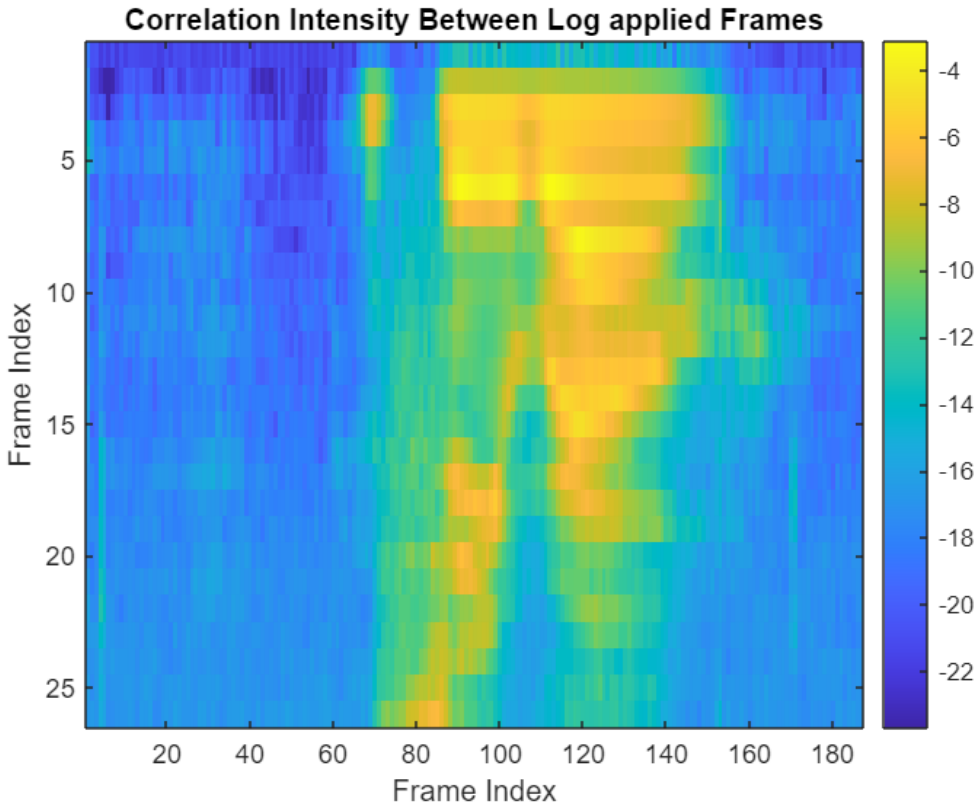
```
% Plot as an image
imagesc(log_mel);

% Adjust axes
axis xy;   % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('Mel Filter Index');
title('Mel Spectrogram (Energy in dB) after applying log');
```

```
colorbar;
```




```
imagesc(log_mel);
colorbar;
xlabel('Frame Index');
ylabel('Frame Index');
title('Correlation Intensity Between Log applied Frames');
```



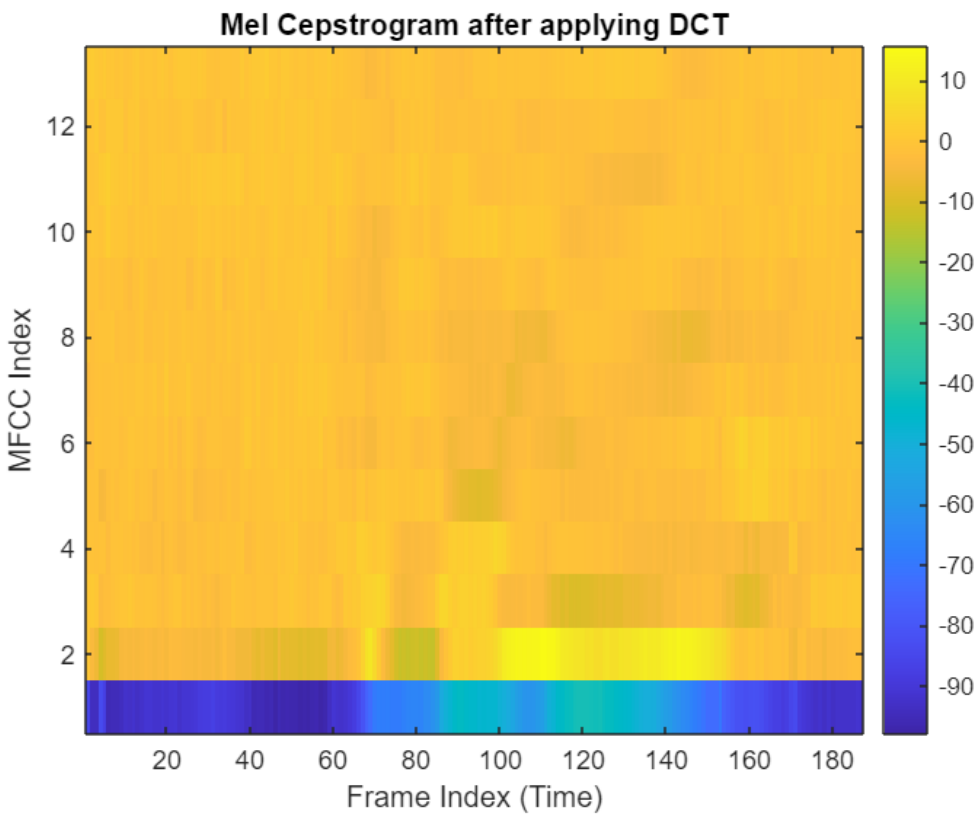
Apply Fourier Cosine transform

```
num_coeffs =13;
mfcc = apply_dct(log_mel, num_coeffs)
```

```
mfcc = 13×187
  -84.6655  -94.0774  -94.1807  -83.6803  -87.5592  -95.7119  -94.7470  -94.8710 ···
    0.2524   -4.0828   -5.6821  -10.5412   -8.8929   -7.3801   -7.1709   -6.6387
    0.4443    1.2478    0.1951   -3.2622   -2.2999   -2.2646   -1.3726   -0.7359
   -0.7725   -0.9594   -1.4289   -0.8312   -0.9034   -2.5740   -1.1433   -1.0117
   -1.0688   -1.4692   -0.9371   -4.5476   -4.0387   -1.9363   -1.1130   -1.6810
   -2.5284   -2.2852   -0.5829   -0.9631   -1.5975   -1.2042   -1.0631   -0.9478
   -2.9888   -1.8545   -1.2236   -0.9473   -1.2335   -1.6290   -0.7763   -0.7465
   -0.5658   -1.8575   -0.9534   -0.5775   -0.4675   -0.7546   -0.2850   -1.6506
   -0.8449   -1.1689   -0.4859   -1.7329   -1.3765   -0.6841   -1.5468   -1.1620
   -0.7725    0.5384    0.0419   -0.4310    0.2572    2.1672    0.5291    0.0586
      ⋮
```

```
% Plot as an image
imagesc(mfcc)
% Adjust axes
axis xy;   % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('MFCC Index');
title('Mel Cepstrogram after applying DCT');
```

```
colorbar;
```

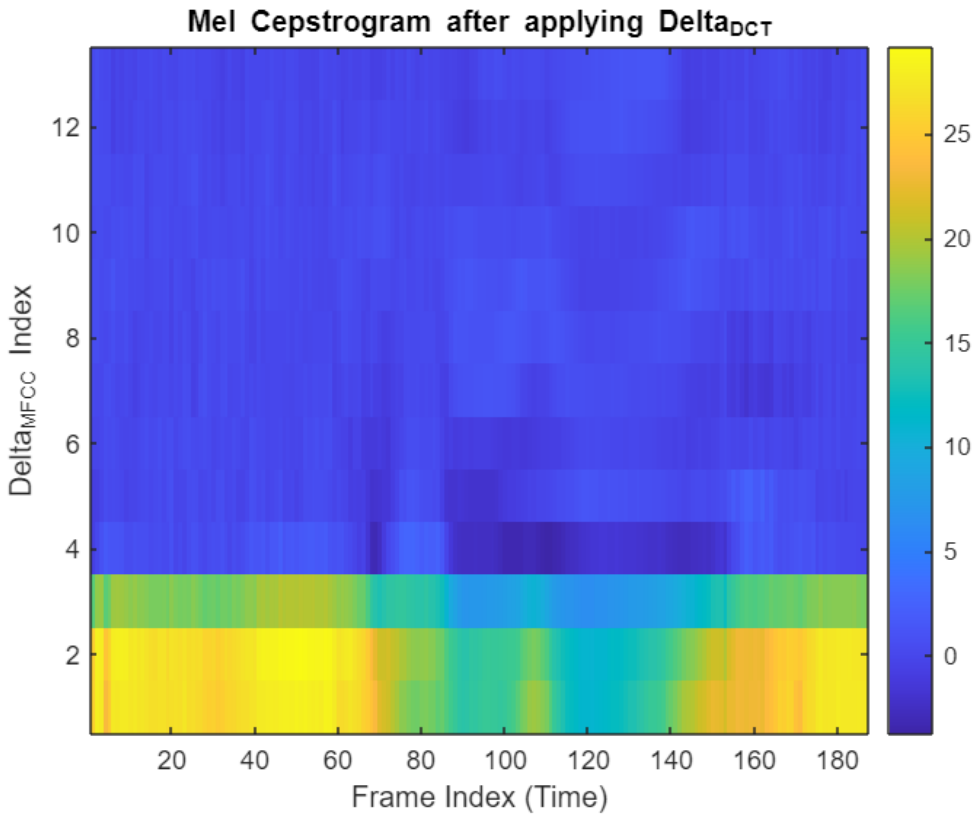


Apply Delta Fourier Cosine transform

```
delta_mfcc = compute_delta(mfcc)
```

```
delta_mfcc = 13×187
   25.5138   28.0645   27.7250   23.3975   24.9185   27.5226   27.4325   27.6503 ···
   25.2896   28.1561   27.9879   24.6116   25.8571   27.9723   28.0582   28.1854
   16.6169   18.8340   19.0740   16.7975   17.5031   19.2357   19.3296   19.2007
   -0.7075    0.0878    0.9066    1.7871    1.2852    1.2680    1.2475    1.0437
   -0.8622   -0.7531   -0.1991    0.4498    0.1439    0.2641    0.1273    0.0043
   -0.1507   -0.2182    0.0665    0.4108    0.3677    0.3946    0.2053   -0.0343
    0.2410    0.1028    0.0532    0.6015    0.6454    0.2954   -0.0090    0.0335
    0.5656    0.6333    0.1987    0.0279    0.3566    0.7688    0.2414    0.1597
    0.5990    0.5216    0.2986    0.4215    0.6283    1.1453    0.4783    0.5033
    0.2685    0.4383    0.4214    0.2347    0.2060    0.6235    0.4522    0.6604
      ⋮
```

```
% Plot as an image
imagesc(delta_mfcc)
% Adjust axes
axis xy;   % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('Delta_MFCC Index');
title('Mel Cepstrogram after applying Delta_DCT');
colorbar;
```

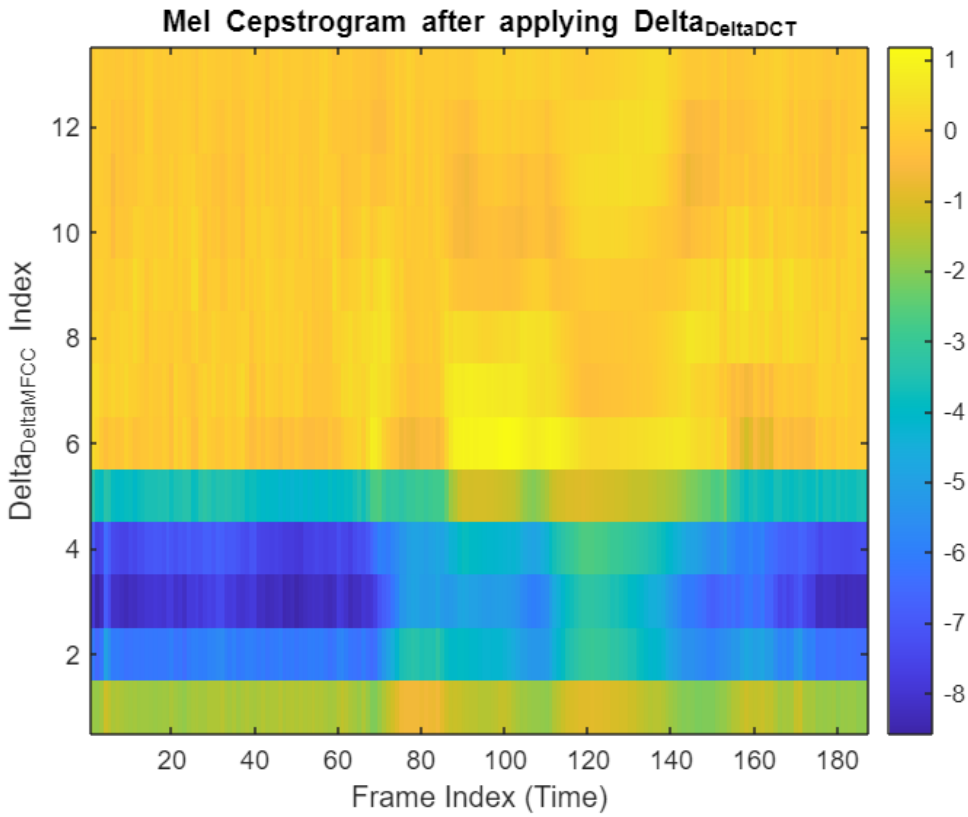


Apply Delta Delta Fourier Cosine transform

```
delta_delta_mfcc = compute_delta(delta_mfcc)
```

```
delta_delta_mfcc = 13×187
   -1.8018   -1.8369   -1.7039   -1.1986   -1.3892   -1.6124   -1.5580   -1.6364 ···
   -6.1339   -6.5184   -6.2288   -4.9821   -5.4682   -6.0796   -6.0473   -6.1663
   -7.8749   -8.5703   -8.2930   -6.8720   -7.4121   -8.1221   -8.1421   -8.2434
   -6.8360   -7.6336   -7.5116   -6.4749   -6.8338   -7.4127   -7.4908   -7.5636
   -3.2195   -3.7768   -3.8882   -3.3768   -3.4633   -3.8754   -3.9719   -3.9412
    0.3649    0.1947   -0.1163   -0.3367   -0.1356   -0.0967   -0.2148   -0.1739
    0.3639    0.3401    0.1128   -0.0440    0.0958    0.2137    0.0738    0.1192
    0.1196    0.1732    0.0955   -0.0532   -0.0340    0.1308    0.0981    0.1859
    0.0284    0.0705    0.0902   -0.0163   -0.0639   -0.0708    0.0989    0.0764
   -0.0145   -0.0234    0.0318    0.0219   -0.0627   -0.3992   -0.0843   -0.1233
      ⋮
```

```
% Plot as an image
imagesc(delta_delta_mfcc)
% Adjust axes
axis xy;   % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('Delta_Delta_MFCC Index');
title('Mel Cepstrogram after applying Delta_Delta_DCT');
colorbar;
```

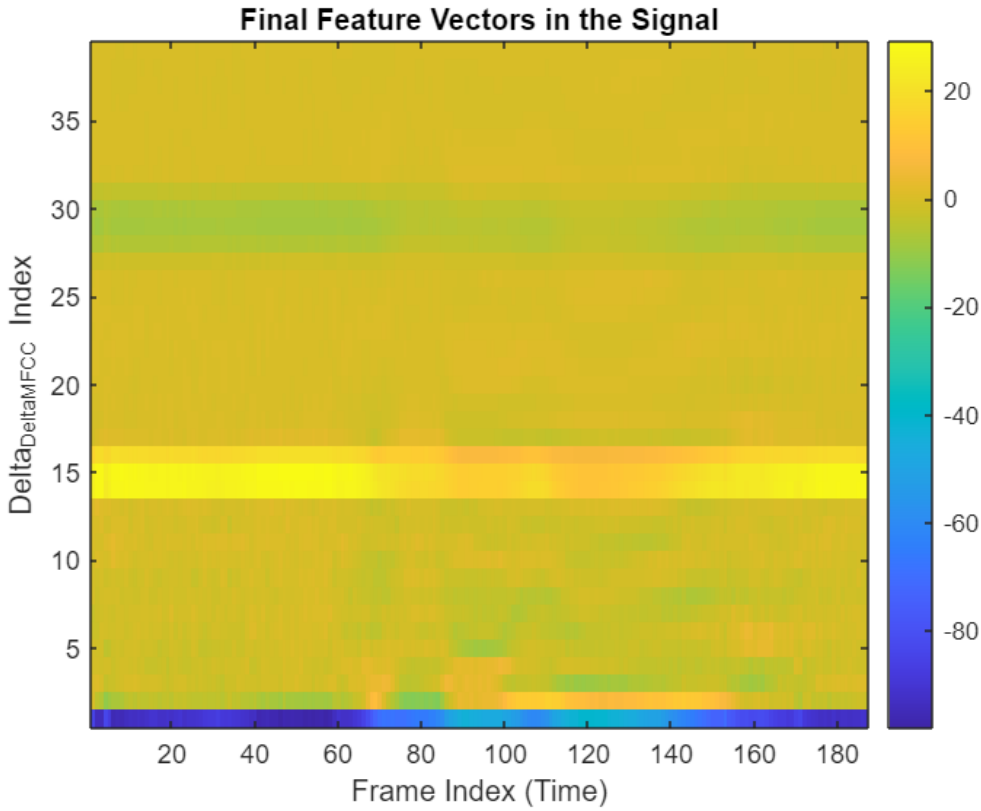


Final Feature Matrix

```
Final_feature_vector=[mfcc;delta_mfcc;delta_delta_mfcc]
```

```
Final_feature_vector = 39×187
  -84.6655  -94.0774  -94.1807  -83.6803  -87.5592  -95.7119  -94.7470  -94.8710 ···
    0.2524   -4.0828   -5.6821  -10.5412   -8.8929   -7.3801   -7.1709   -6.6387
    0.4443    1.2478    0.1951   -3.2622   -2.2999   -2.2646   -1.3726   -0.7359
   -0.7725   -0.9594   -1.4289   -0.8312   -0.9034   -2.5740   -1.1433   -1.0117
   -1.0688   -1.4692   -0.9371   -4.5476   -4.0387   -1.9363   -1.1130   -1.6810
   -2.5284   -2.2852   -0.5829   -0.9631   -1.5975   -1.2042   -1.0631   -0.9478
   -2.9888   -1.8545   -1.2236   -0.9473   -1.2335   -1.6290   -0.7763   -0.7465
   -0.5658   -1.8575   -0.9534   -0.5775   -0.4675   -0.7546   -0.2850   -1.6506
   -0.8449   -1.1689   -0.4859   -1.7329   -1.3765   -0.6841   -1.5468   -1.1620
   -0.7725    0.5384    0.0419   -0.4310    0.2572    2.1672    0.5291    0.0586
      ⋮
```

```
% Plot as an image
imagesc(Final_feature_vector)
% Adjust axes
axis xy;   % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('Delta_Delta_MFCC Index');
title('Final Feature Vectors in the Signal');
colorbar;
```


Final Feature Vectors in the Signal
Delta$_{\text{DeltaMFCC}}$ Index
35
30
25
20
15
10
5
20
40
60
80
100
120
140
160
180
Frame Index (Time)
20
0
-20
-40
-60
-80